Optimization

2.4 Bisection method

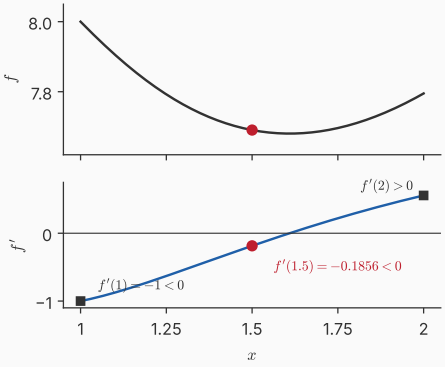
José Firmino Aguilar Madeira

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Can we do the same using only the sign of f'?

Guiding example: $f(x) = 8e^{1-x} + 7\ln x$ on $[1, 2]$, $f'(x) = -8e^{1-x} + 7/x$. A single point: $f'(1.5) = -8e^{-0.5} + 7/1.5 = -0.1856 < 0$. At 1.5, f is still *decreasing*, so $x^* > 1.5$.



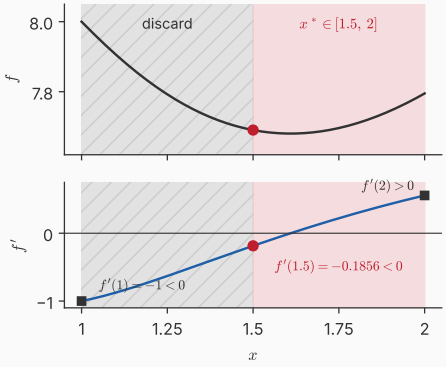
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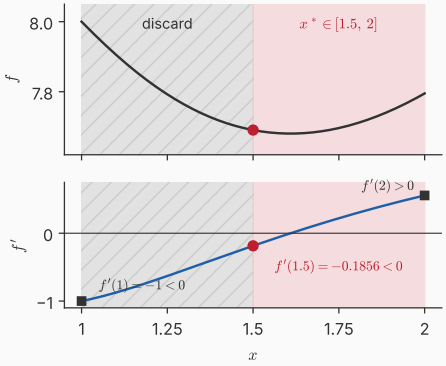
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Where to put the point? In the middle: whatever the sign, half remains.

sign of f' at the midpoint $\rightarrow$ **bisection**



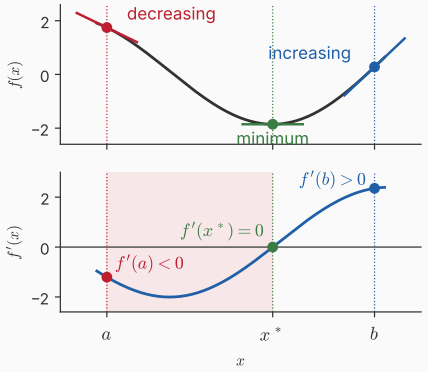
Why the sign is enough

Assumptions: f' continuous on $[a, b]$, $f'(a) < 0 < f'(b)$ and a single zero, where f' changes sign $- \to +$. Then that zero is the only interior minimum x^*. In particular, f strictly convex with $f'(a) < 0 < f'(b)$ is enough.

The sign of f' at a point tells on which side the minimum is:

- $f'(x) < 0$: f is still decreasing $\Rightarrow x^* > x$;
- $f'(x) > 0$: f is already increasing $\Rightarrow x^* < x$.

It tells *on which side*, not *where*:
$f' = -0.19$ and $f' = -0.0001$: the same half.



$f'(a) < 0 < f'(b)$ alone guarantees *at least* one zero; without uniqueness there may be several stationary points and we do not know whether we isolate the desired minimum.

The algorithm

Bisection (on f')

Step 1 Choose $a < b$ with $f'(a) < 0 < f'(b)$ and $\varepsilon > 0$; set $x_1 = a$ and $x_2 = b$.

Step 2 Set $z = \dfrac{x_1 + x_2}{2}$ and compute $f'(z)$.

Step 3 If $|f'(z)| \leq \varepsilon$, terminate: $x^* \approx z$. Otherwise, if $f'(z) < 0$, set $x_1 = z$ (minimum to the right); if $f'(z) > 0$, set $x_2 = z$ (minimum to the left). Go to Step 2.

Note: $|f'(z)| \leq \varepsilon$ says that f is nearly flat at z, *not* that z is within ε of x^*. Variant: stop when $x_2 - x_1 \leq \varepsilon_x$ and return the midpoint, which guarantees $|x - x^*| \leq \varepsilon_x/2$.

- Order 1 information: it uses f' (its sign, to choose the half; its value, to stop), not f.
- It requires $f'(a) < 0 < f'(b)$: check before starting. If it fails, the minimum is not bracketed and bisection does not apply in this form.
- Cost: one evaluation of f' per iteration. If f' is not available analytically, it is approximated by finite differences (2.6) — two evaluations of f.

How many iterations? And how much does each one cost?

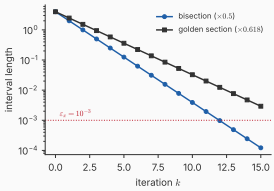
Each iteration halves the interval: after k iterations, $x_2 - x_1 = \left(\frac{1}{2}\right)^k (b-a)$. With $|f'(z)| \le \varepsilon$, the number of iterations depends on f and is not known in advance. In the variant (stop when $x_2 - x_1 \le \varepsilon_x$; the midpoint has $|x - x^*| \le \varepsilon_x/2$), it is:

$$\boxed{k = \left\lceil \log_2 \frac{b-a}{\varepsilon_x} \right\rceil}$$

Example

$[a,b] = [0,2]$, $\varepsilon_x = 10^{-3}$: $\log_2 2000 = 10.97$, so $k = 11$. With $\varepsilon_x = 10^{-4}$: $k = 15$.

Each decimal digit costs $\log_2 10 \approx 3.3$ iterations (golden section: 4.8).



But what is the cost per evaluation of f?

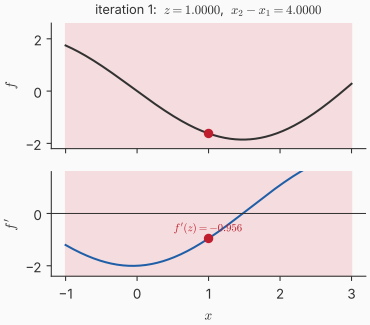
Bisection: $\times 0.5$ per f'; golden section: $\times 0.618$ per new f. With f' by central differences (2 evaluations of f): $\sqrt{0.5} \approx 0.707$ per evaluation — worse.

A cost conversion, not an intrinsic rate of bisection; it assumes 2 new evaluations of f per f', with no reuse.

Example 1: $f(x) = -2\sin x + x^2/16$ on $[-1, 3]$

$f'(x) = -2\cos x + x/8$; $\quad f'(-1) = -1.21 < 0$,
$f'(3) = 2.35 > 0$ ✓; $\quad \varepsilon = 0.03$

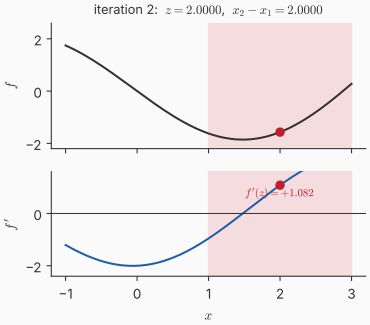
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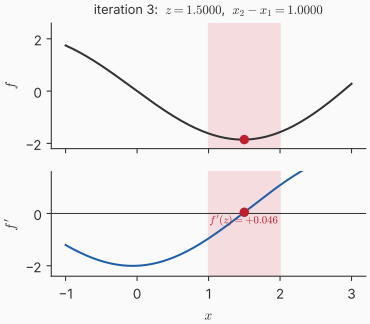
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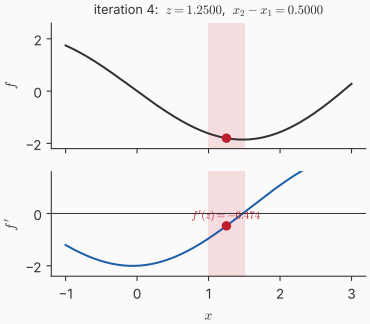
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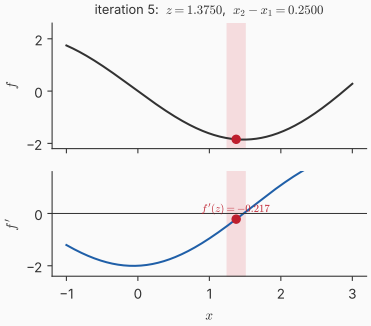
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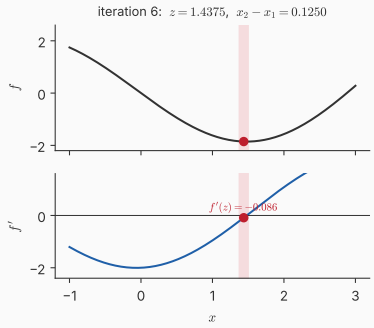
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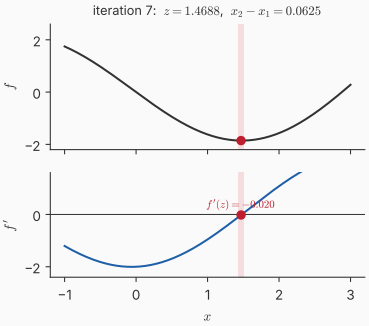


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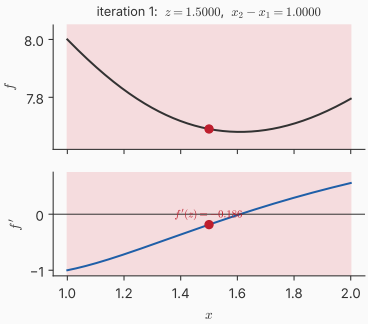
$|f'(z)| = 0.0201 \le \varepsilon$: terminate, $x^* \approx z = 1.4688$ (exact: 1.4783).
With $\varepsilon = 0.05$ it would terminate at iteration 3, at $z = 1.5$ ($|f'(z)| = 0.0460$), with error 0.022.



Example 2: $f(x) = 8e^{1-x} + 7\ln x$ on $[1, 2]$ (2019 first test)

$f'(x) = -8e^{1-x} + 7/x$; $\quad f'(1) = -1 < 0$, $f'(2) = 0.557 > 0$ ✓; $\quad \varepsilon = 0.01$ (the golden-section interval of 2.2)

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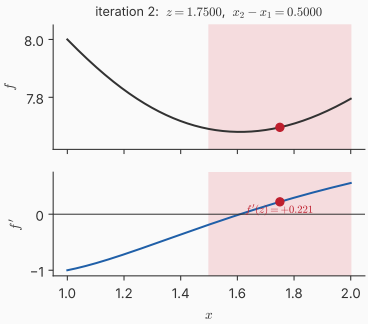


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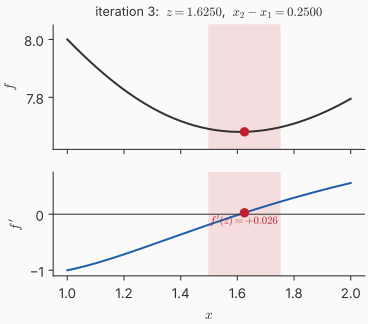
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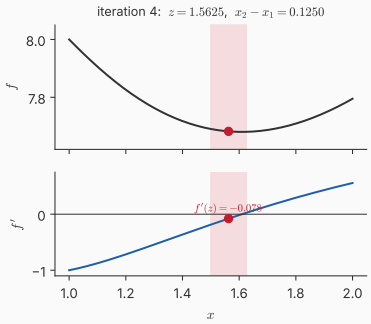
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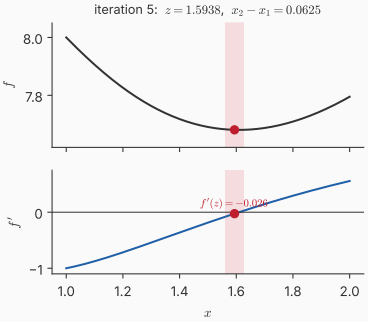
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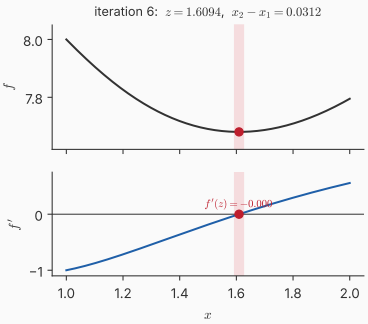


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5	1.5625	1.6250	1.5938	7.68065	−0.02586
6	1.5938	1.6250	1.6094	7.68045	−0.00001

$|f'(z)| = 0.00001 \le \varepsilon$: terminate, $x^* \approx z = 1.6094$ (exact 1.60938; error $6 \cdot 10^{-6}$, by luck: the interval guarantees only 0.016). With $\varepsilon = 0.03$ it would stop at it. 3, at $z = 1.625$. Variant with 15 reductions: $x_2 - x_1 = 1/2^{15} = 3.1 \times 10^{-5}$; golden section, 15 reductions: $L = 0.0007$, 24 times larger (but with f only). The method does not use the $f(z)$ column.



To solve

Apply bisection with $\varepsilon = 0.01$ and confirm:

$f(x)$	interval	iterations	final z	$f(z)$
$x + \dfrac{31}{x+3}$	$[0, 5]$	6	2.5781	8.1355
$-3\sin x + \dfrac{x^2}{8}$	$[0, 4]$	10	1.4492	-2.7153

- In the first, check analytically: $f'(x) = 1 - \dfrac{31}{(x+3)^2} = 0 \Rightarrow x^* = \sqrt{31} - 3 = 2.5678$. The error of z is $0.0104 > \varepsilon$: $|f'(z)| \leq \varepsilon$ does not guarantee $|z - x^*| \leq \varepsilon$.
- In the second ($x^* = 1.4497$), compare with golden section in 2.2 (1.4489, $L_{15} = 0.0029$) and with the variant $x_2 - x_1 \leq \varepsilon_x$ with 15 reductions ($4/2^{15} = 1.2 \times 10^{-4}$): after 15 reductions, which one guarantees the smaller error?

On the computer

The core, in MATLAB (`tol` $= \varepsilon$):

```
if ~(df(a)<0 && df(b)>0), error('bracketing'), end
x1 = a;  x2 = b;
while true
    z = (x1 + x2)/2;  d = df(z);
    if abs(d) <= tol, break, end  % finished
    if d < 0
        x1 = z;      % minimum to the right
    else
        x2 = z;      % minimum to the left
    end
end
x = z;
```

The full function (`Bisection`) is in the course companion code.

It uses only f': $n_g = 1$ per iteration, plus $f'(a)$ and $f'(b)$ to check the bracketing ($n_g = k + 2$); $n_f = 0$.
The full function also has a maximum number of iterations and, optionally, the variant $x_2 - x_1 \leq \varepsilon_x$.

Without an analytical f', central differences (2.6), $n_f = 2$ per f':

```
h  = 1e-6;
df = @(x) (f(x+h) - f(x-h)) / (2*h);
```

near the zero, the sign of f' may also have numerical noise — see 2.6

KEY TAKEAWAYS — 2.4

1. For an interior minimum of a differentiable f, $f'(x^*) = 0$; under the assumptions of this section, minimizing is equivalent to locating that zero.
2. With f' continuous, $f'(a) < 0 < f'(b)$ and a single zero, the sign at the midpoint tells which half to keep: $f'(z) < 0 \Rightarrow x_1 = z$; $f'(z) > 0 \Rightarrow x_2 = z$. Terminate when $|f'(z)| \leq \varepsilon$.
3. $L_k = 2^{-k} L_0$: linear convergence with ratio $\frac{1}{2}$; in the variant $x_2 - x_1 \leq \varepsilon_x$, the number of iterations is predictable, $k = \lceil \log_2(L_0/\varepsilon_x) \rceil$. One evaluation of f' per iteration, none of f (with f' by central differences, golden section reduces more per evaluation of f).
4. It chooses the half by the sign of f' only: it knows which side the zero is on, not where.

Next (2.5): what if we also use the *value* of $f'(z)$ and the slope of f', that is $f''(z)$, to estimate directly where the zero is? — the Newton–Raphson method.

References and resources

References Full list: *Course bibliography*, at the end of the course PDF.

- Deb, K., *Optimization for Engineering Design: Algorithms and Examples*, 2nd ed., PHI Learning, 2012 — Ch. 2.
- Dennis, J. E., Jr., Schnabel, R. B., *Numerical Methods for Unconstrained Optimization and Nonlinear Equations*, SIAM (Classics in Applied Mathematics 16), 1996 — Ch. 2.

Resources

- Companion code: `Bisection` (stopping rule $|f'(z)| \leq \varepsilon$; variant $x_2 - x_1 \leq \varepsilon_x$); f' by central differences (`df`, $h = 10^{-6}$).